# BUSINESS FINANCE — FINANCIAL RATIOS

Chapter 3 | One-page formula sheet · Use figures from the same fiscal period where possible

## 1. COMMON-SIZE STATEMENTS

| Ratio / purpose | Formula |
|---|---|
| Balance sheet account (%) | Account ÷ Total Assets × 100 |
| Income statement item (%) | Item ÷ Sales × 100 |

## 2. LIQUIDITY — short-term solvency

| Ratio / purpose | Formula |
|---|---|
| Current ratio | Current Assets (CA) ÷ Current Liabilities (CL) |
| Quick ratio | (CA – Inventory) ÷ CL |
| Cash ratio | Cash ÷ CL |

## 3. LEVERAGE / LONG-TERM SOLVENCY

| Ratio / purpose | Formula |
|---|---|
| Total debt ratio | (Total Assets – Total Equity) ÷ Total Assets = Total Debt ÷ TA |
| Debt-to-equity (D/E) | Total Debt ÷ Total Equity |
| Equity multiplier (EM) | Total Assets ÷ Total Equity = 1 + D/E |
| Times interest earned (TIE) | EBIT ÷ Interest |
| Cash coverage | (EBIT + Depreciation) ÷ Interest |

## 4. ASSET MANAGEMENT / TURNOVER

| Ratio / purpose | Formula |
|---|---|
| Inventory turnover | COGS ÷ Inventory |
| Receivables turnover | Sales ÷ Accounts Receivable (AR) |
| Payables turnover | COGS ÷ Accounts Payable (AP) |
| Total asset turnover (TAT) | Sales ÷ Total Assets |

## 5. PROFITABILITY

| Ratio / purpose | Formula |
|---|---|
| Profit margin (PM) | Net Income (NI) ÷ Sales |
| Return on assets (ROA) | NI ÷ Total Assets |
| Return on equity (ROE) | NI ÷ Total Equity |

## 6. MARKET VALUE

| Ratio / purpose | Formula |
|---|---|
| Earnings per share (EPS) | Net Income ÷ Shares Outstanding |
| P/E ratio | Price per Share ÷ EPS |
| Price/Sales (P/S) | Price per Share ÷ Sales per Share |
| Market-to-book (M/B) | Price per Share ÷ Book Value per Share |
| Enterprise value (EV) | Market Value of Equity + Book Value of Liabilities – Cash |
| EBITDA | EBIT + Depreciation & Amortization |
| EV/EBITDA | Enterprise Value ÷ EBITDA |

## 7. DUPONT IDENTITY — decompose ROE

| Ratio / purpose | Formula |
|---|---|
| Core identity | ROE = PM × TAT × EM |
| Expanded | (NI ÷ Sales) × (Sales ÷ TA) × (TA ÷ Equity) = NI ÷ Equity |

## QUICK INTERPRETATION & EXAM CHECKS

**Liquidity:** higher ratios usually mean more short-term coverage, but excess idle assets may reduce efficiency. • **Leverage:** more debt can magnify shareholder returns and risk; TIE / cash coverage measure ability to cover interest. • **Turnover:** higher turnover generally means more efficient asset use; interpret inventory turnover in industry context. • **Profitability:** PM = profit per sales dollar; ROA = profit relative to assets; ROE = profit relative to equity. • **Common-size:** balance sheet uses total assets as denominator; income statement uses sales. • **DuPont:** ROE can come from high margin, efficient asset turnover, or high leverage. • **Calculation:** convert percentages to % only at the end; ratios/turnover are often shown as “x” (e.g., 1.5x). • **Comparison cautions:** compare across time and similar firms; accounting methods, seasonality, fiscal year-end, one-offs, and window dressing can mislead.